

O2MECH - TEST 2 - SOLUTIONS

① $\vec{r}_1 = (at^2, vt, 0)$ $\vec{r}_2 = (vt, at^2, b)$ $\vec{r}_3 = (0, b, at^2)$
 $\vec{v}_1 = (2at, v, 0)$ $\vec{v}_2 = (v, 2at, 0)$ $\vec{v}_3 = (0, 0, 2at)$
 $\vec{a}_1 = (2a, 0, 0)$ $\vec{a}_2 = (0, 2a, 0)$ $\vec{a}_3 = (0, 0, 2a)$
 $\vec{F}_{tot} = \sum_{i=1}^3 m \vec{a}_i = (2am, 2am, 2am)$: Linear momentum is not conserved $\frac{d\vec{p}}{dt} = \vec{F}_{tot} \neq 0$
 $\Rightarrow \vec{p}$: constant
 $\vec{N}_{tot} = \sum_{i=1}^3 \vec{r}_i \times \vec{F}_i = m(at^2, vt, 0) \times (2a, 0, 0) + m(vt, at^2, b) \times (0, 2a, 0) + m(0, b, at^2) \times (0, 0, 2a)$
 $= m(0, 0, -2avt) + m(-2ab, 0, 2avt) + m(2ab, 0, 0)$
 $= m(0, 0, 0)$: Angular momentum is conserved $\frac{d\vec{L}}{dt} = \vec{N}_{tot} = 0$
 $\Rightarrow \vec{L}$: constant.

② $e^{-\beta t} x_0 = \frac{x_0}{2} \rightarrow \ln e^{-\beta t} = \ln 2^{-1} \rightarrow -\beta t = -\ln 2 \rightarrow \frac{\beta t}{2m} = \ln 2 \rightarrow t = \frac{2m \ln 2}{b}$
 $\Rightarrow t = \frac{2 \times 0.7}{1.4} = 1s$

③ a) $\frac{1}{r} = \frac{1 + \epsilon \cos \theta}{\alpha}$, $\frac{d}{d\theta} \left(\frac{1}{r} \right) = \frac{-\epsilon \sin \theta}{\alpha}$, $\frac{d^2}{d\theta^2} \left(\frac{1}{r} \right) = \frac{-\epsilon \cos \theta}{\alpha}$
 $-\frac{\epsilon \cos \theta}{\alpha} + \frac{1}{r} = -\frac{\mu}{\ell^2} r^2 f(r) \rightarrow -\frac{\epsilon \cos \theta}{\alpha} + \frac{1 + \epsilon \cos \theta}{\alpha} = -\frac{\mu}{\ell^2} r^2 f(r)$
 $\Rightarrow \frac{1}{\alpha} = -\frac{\mu}{\ell^2} r^2 f(r) \rightarrow f(r) = -\frac{\ell^2}{\alpha \mu r^2} = -\frac{\frac{\ell^2}{\mu k}}{\mu r^2} = -\frac{k}{r^2}$ attractive inverse-square law.

b) $f(r) = -\frac{k}{r^2} \Rightarrow \vec{F} = -\vec{\nabla} U \Rightarrow f(r) = -\frac{dU}{dr} \Rightarrow U(r) = -\int f(r) dr + C \Rightarrow U(r) = k/r$

④ a) $\vec{F}_c = -2m(\omega_z \hat{e}_3) \times (V_0 \hat{e}_2) = 2m\omega_z V_0 \hat{e}_1$

b) The particle has a constant velocity in $\hat{e}_2$ direction with magnitude V_0

If also has an acceleration in $\hat{e}_1$ direction because of $\vec{F}_c$.

In $\hat{e}_2$ direction: $x_2(t) = V_0 t$

In $\hat{e}_1$ direction: ~~Newton's 2nd law~~ $2m\omega_z V_0 = m\ddot{x}_1$

initial velocity in $\hat{e}_1$ direction is zero.

$\dot{x}_1(t) \equiv \dot{x}_2(t) = 2\omega_z V_0 t$

$x_1(t) = \omega_z V_0 t^2 + x_1(0)$ we can also assume that initial position of the particle is zero

$$\begin{aligned} x_1(t) &= \omega_z V_0 t^2 \\ x_2(t) &= V_0 t \end{aligned}$$